MANIPAL UNIVERSITY JAIPUR, JAIPUR
B.Tech First Year
BLUEPRINT END TERM EXAMINATION – July-Nov 2026
Problem Solving Using Computers (PSUC) – CSC1101

Syllabus:

Introduction to assembly and high-level programming languages (programming paradigms), Number System, Basic concepts of integer and floating-point number representations;

Fundamentals of Data Types and Variables in Programming: Data types, constants, variables, enumerated data types, Storage Class
Understanding Operators, Expressions, and Control Statements: Operators, expressions, Control Statements and branch statements, iteration statements
Working with Arrays -1D, 2D and Pointers in Programming: Arrays and pointers, Strings
Functions and parameter passing, Recursion, Structures in Programming: Structure – Nested structures, Array of structures, Union
Standard Library Functions, and Elementary Data Structures

Module	Syllabus Contents	Marks
Module -1	Von Neumann Architecture of Computer, Introduction to assembly and high-level programming languages (programming paradigms), Number System, Basic concepts of integer and floating-point number representations; Basic Structure of C program and Preprocessor Directives, Macros, Fundamentals of Data Types and Variables in Programming: Data types, constants, variables, Enumerated data types, Storage Classes, Operators in C, Type Casting and Type Conversion, Operator Precedence and Associativity, Format Specifiers in C	12-14 Marks
Module -2	Conditional and branch statements-if, if-else, nested if-else, if-else ladder, Switch statement, break, continue, and goto statements, Iteration in C - while, do-while, and for loop	16-20 Marks
Module -3	1-D Array, 2-D Array, 1-D Character strings, Linear Search and Bubble Sort	06-10 Marks
Module -4	Functions, Call by Value and Call by Reference, Recursion, Pointers, Pointers and Arrays.	24-28 Marks
Module -5	Structures, nested structures, arrays of structures, and Union	10-12 Marks